

Chapter 80 — Surgery of Testicular Tumours — Numbers, Gold Standards and Traps

Abbreviations: AFP α -fetoprotein · β -hCG β -human chorionic gonadotropin · BEP bleomycin, etoposide, cisplatin · CR complete response · CS clinical stage · CSM cancer-specific mortality · CSS cancer-specific survival · CT computed tomography · DVT deep vein thrombosis · EAU European Association of Urology · EP etoposide, cisplatin · FDG-PET fluorodeoxyglucose positron emission tomography · FiO₂ fraction of inspired oxygen · GCT germ cell tumour · HDCT high-dose chemotherapy · HPF high-power fields · IGCCCG International Germ Cell Consensus Classification Group · IMA/IMV inferior mesenteric artery/vein · ITGCN intratubular germ cell neoplasia · IVC inferior vena cava · LDH lactate dehydrogenase · LVI lymphovascular invasion · MCT medium-chain triglycerides · MI myocardial infarction · NCCN National Comprehensive Cancer Network · NG nasogastric · NPO nil by mouth · NPV/PPV negative/positive predictive value · NSGCT nonseminomatous GCT · OS overall survival · PC post-chemotherapy · PE pulmonary embolism · PFS progression-free survival · PFTs pulmonary function tests · pN pathological nodal stage · RCT randomised controlled trial · RFS recurrence-free survival · RP retroperitoneal · RPLND retroperitoneal lymph node dissection · SCST sex cord stromal tumour · SCT stem cell transplantation · SEMPET seminoma PET study · SpO₂ oxygen saturation · TG triglyceride · TPN total parenteral nutrition · VTE venous thromboembolism

IF YOU READ NOTHING ELSE

- **Suspected cancer** → **inguinal radical orchidectomy**, early cord control, cord to internal ring
- **Partial orchidectomy only** if polar + ≤ 2 cm + contralateral testis compromised/absent
- **PC-RPLND**: nonseminoma residual >1 cm + normal markers
- **Complete response** (≤ 1 cm, markers normal) → **observe**
- **PC-RPLND standard** = residual mass + **full bilateral inguinal template**
- **Modified PC template**: ≤ 5 cm in landing zone + normal markers + good/intermediate risk
- **PC-RPLND histology**: fibrosis 40% · teratoma 45% · viable 15%
- **Seminoma residual**: <3 cm observe; >3 cm → FDG-PET; PET-avid → treat
- **Late relapse** (>2 years) resectable → **surgery first** (chemoresistant)
- **Bleomycin** → low FiO₂ (SpO₂ 88–90%), restrictive fluids

MANAGEMENT OF THE TESTIS MASS

- **Ultrasound key role**: contralateral testis; **bilateral synchronous** $\approx 1\%$
- **Markers**: AFP, β -hCG, LDH before and after orchidectomy; trend guides
- **Staging**: CT with IV + oral contrast; **no role for PET/lymphangiography** initially
- **Incision**: 3–5 cm inguinal, Langer lines; **preserve ilioinguinal nerve**
- **Vas and vessels tied separately**; non-absorbable, **1–2 cm tail** for RPLND
- **Partial**: hypothermia unnecessary if <30 min
- **Tunica incision**: vertical ventral midline vs horizontal medial/lateral (segmental arteries)
- **Partial adjuvant 18–20 Gy** → spermatogenesis lost, Leydig function kept
- **Mass <5 mm, markers negative** → partial or surveillance
- **Normal contralateral testis** → **no elective testis sparing**
- **Local recurrence after partial** → **completion orchidectomy**, always
- **Metastatic presentation** → chemotherapy first → **delayed orchidectomy in all**
- **Extragenital primary**: ITGCN up to 42%; metachronous $\approx 5\%$; lateralising → orchidectomy

RETROPERITONEAL LYMPH NODE DISSECTION

- **Primary RPLND**: high-risk CS I or CS IIA (N1) NSGCT, normal markers
- **Late relapse** = >24 months after complete response
- **Pre-op**: CT ≤ 6 weeks; markers ≤ 7 –10 days; **no bowel prep**; offer sperm banking
- **Bleomycin lung toxicity 11% vs 1.7%**; grade 3–4 $\times 5$ → PFTs
- **Exposure**: caecum → IMV; Toltd → foramen of Winslow
- **Split at 12 o'clock on aorta** below left renal vein → IMA
- **Lumbar arteries: 3 pairs; 2nd pair at IMA level**
- **Common lumbar vein**: largest, at IMA, may pass behind aorta
- **Sympathetic L1–L4** → anterior aorta → superior hypogastric plexus

- **Right sympathetic trunk** posterolateral to IVC — not lymphatic; preserve
- **Drain** only for large resections: non-suction; out when serous <100 mL/24 h
- **Post-op**: no NG tube; fluids day 0; walk day 1; **discharge day 3–5**

AUXILIARY PROCEDURES

- **Auxiliary in 23–45% of PC-RPLND**; commonest **nephrectomy (5–31%)** > vascular
- **Nephrectomy predictor**: residual >10 cm, especially left-sided
- **IVC intervention**: mass >5 cm $\approx 20\%$ vs <5 cm $\approx 3\%$
- **IVC narrowing $<25\%$ → primary repair**; $>25\%$ → patch or graft
- **IVC ligation acceptable**; significant **aortic injury → graft always**
- **Aortic replacement $\approx 1\%$** ; long duodenal serosal injury → graft (aortoduodenal fistula)
- **Intraluminal thrombus $\approx 6\%$** (IVC/renal vein)
- **IGCCCG**: non-pulmonary visceral metastases = **poor risk** ($\approx 48\%$ 5-year)
- **Liver discordance $\approx 50\%$** ; RP necrosis → liver necrosis 94%
- **Unresectable liver lesion <3 cm** → open microwave ablation
- **Supradiaphragmatic 10–20%**; 80% lower/middle mediastinum
- **Anterior mediastinum** → think primary mediastinal GCT
- **Thoracic discordance 25–50%**; RP necrosis → chest necrosis 89% → **RPLND first**
- **Resect thoracic residual >1 cm** unless RP fibrosis only
- **Retrocrural**: transdiaphragmatic; **$>T8$ → partial sternotomy**; extensive chest → staged

SURGICAL DECISION MAKING

- **Complete response $\approx 70\%$** after cisplatin (CS $\geq$ II)
- **≤ 1 cm residual**: teratoma $\approx 22\%$, viable $\approx 5\%$
- **Necrosis predictors**: no primary teratoma, seminoma elements, normal AFP/hCG, raised LDH, $>50\%$ shrinkage, <2 cm
- **Einhorn & Donohue 1977**: curative cisplatin
- **Donohue 1982 mapping**: right → interaortocaval/precaval; left → para-aortic
- **Omit suprahilar + interiliac** if non-bulky

- **3–23% nodes outside modified templates** (right > left) → full bilateral argument
- **Primary series:** pN+ 20–30%; relapse pN0 7–12%; pN+ alone 23–34%; CSS ≈99–100%
- **pN1–2 cured by RPLND alone ≈70%**
- **Williams 1987 RCT:** relapse 40/53/60% (micro/<2/>2 cm), not significant
- **Adjuvant:** pN1 observe; pN2 2 cycles vs observe; teratoma only → observe
- **Salvage after relapse:** ≥1 extra BEP or 2 extra EP
- **Modified PC template results:** in-field recurrence 0–4%, CSS 98–100%, ejaculation >85%

HISTOLOGY AND OUTCOMES AFTER CHEMOTHERAPY

- **Fibrosis CSS/RFS ≈95%; teratoma RFS 80–90%, CSS 85–95%; viable 50–70%**
- **Fizazi factors:** incomplete resection · ≥10% viable · IGCCCG intermediate/poor
- **Fizazi 0/1/≥2:** 5-year OS 100/83/51%; PFS 90/76/38%
- **Adjuvant after viable GCT:** ↑PFS, not OS
- **NCCN:** incomplete or intermediate/poor; **EAU:** not good-risk <10% viable
- **No adjuvant** after salvage chemotherapy, fibrosis or teratoma

PC-RPLND IN HIGH-RISK POPULATIONS

- **Salvage:** after ≥7–8 cycles; viable ≥2× first-line; **OS 60–75%**
- **After HDCT:** viable ≈40%; 5-year OS ≈70%
- **Desperation:** 5-year OS ≈54%; worse after salvage, raised β-hCG, redo
- **Plateaued markers** → >75% fibrosis/teratoma
- **Desperation — operate if:** limited, fully resectable, markers falling/plateaued; or chemo exhausted
- **Do not operate:** rising markers, rapid doubling, multisite progression
- **Redo:** incomplete lumbar division → recurrence; ≈half retroaortic/retrocaval
- **Volume:** mean 1/year; >2/year top 25%; 3/year top 10%
- **Late relapse:** 2–4%; RP commonest; yolk sac; OS ≈60%

OUTCOMES, FUNCTION AND COMPLICATIONS

- **Abnormal semen at diagnosis 40–60%**
- **Ejaculation:** template 75–87% → 97%; nerve-sparing 90–100%; paternity ≤75%
- **Complications:** primary 10.6–24% vs PC up to 56%
- **Pulmonary PC 3–5%;** tachycardia is **not a fluid trigger**
- **Ileus up to 26%;** less with extraperitoneal approach
- **Lymphocele** symptomatic ≤1.7%; air/rim → infected → percutaneous drain

- **Chylous ascites:** primary 0.2–2.1% vs PC 2–12%
- **Chyle:** milky, protein >3 g/dL, TG 2–8× serum
- **Cisterna chyli:** L1–L2, behind aorta under left crus; present ≈50%
- **Chyle ladder:** tap → low-fat + MCT + octreotide → NPO/TPN → embolise/shunt/surgery
- **Low-fat diet ≥4 weeks** after chyle resolves
- **VTE:** primary <1%, PC 0.1–3.1%; bleomycin DVT ≈13% → prophylaxis
- **PC mortality <1%:** PE, duodenal fistula, MI, chyle
- **Paraplegia risk:** bulky post-chemo mediastinal/RP disease

RPLND IN UNIQUE SITUATIONS

- **Seminoma CR 70–90%;** viable residual ≈10%; no teratoma
- **3 cm rule:** <3 cm all fibrosis; >3 cm ≈30% viable/teratoma
- **SEMPET >3 cm:** specificity 100%, sensitivity 80%, PPV 100%, NPV 96%
- **SEMPET ≤3 cm:** PPV 37% → PET useless
- **PET-avid >3 cm** → PC-RPLND, salvage chemo or HDCT + SCT
- **Residual seminoma at PC-RPLND:** 5-year CSS ≈54%
- **HDCT seminoma OS:** 92% second-line; 64% third/fourth
- **SCST <5%;** malignant if ≥2: age, >4 cm, necrosis, mitoses >3–5/10 HPF, atypia, infiltration, LVI
- **Sertoli:** 35% stage II/III, CSM 32% vs **Leydig:** 98.5% stage I, CSM 2%
- **RPLND in metastatic SCST:** no proven survival benefit
- **Overall cure >90%**

CLASSIC TRAPS

- **Trap:** scrotal biopsy/approach — **Actually:** inguinal orchiectomy, early cord control
- **Trap:** partial with normal contralateral testis — **Actually:** never elective
- **Trap:** PET for seminoma <3 cm — **Actually:** only >3 cm
- **Trap:** adjuvant after teratoma-only RPLND — **Actually:** observe
- **Trap:** Fizazi model guides adjuvant — **Actually:** prognostic only
- **Trap:** modified template standard post-chemo — **Actually:** full bilateral
- **Trap:** fibrosis at desperation RPLND reassuring — **Actually:** marker source elsewhere
- **Trap:** ligate injured aorta like IVC — **Actually:** graft
- **Trap:** post-op tachycardia → fluid bolus — **Actually:** restrict (bleomycin)
- **Trap:** late relapse → chemotherapy first — **Actually:** resect first